

Python Quiz Application

An Automated Assessment Solution

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| **Introduction**

A Quiz Application is a digital assessment tool that delivers questions and evaluates performance.

- *Scalable digital testing for modern learning.*
- *Eliminates manual grading errors.*
- *Provides instant, data-driven feedback.*

| Project Overview

This console-based Python application serves as an automated examination interface using Object-Oriented Programming (OOP) principles.

| Objectives

Automation

Remove human proctoring.

Evaluation

Real-time grade calculation.

Accuracy

100% data precision.

| *Key Features*

- *Multiple-Choice Question bank.*
- *Automated percentage-based grading.*
- *Robust input validation.*

| *Technologies Used*

Python 3.x, Functions, Lists, Dictionaries, OOP, Control Structures.

| *System Requirements*

Software

Python 3.x, VS Code / IDLE

Hardware

Dual-core CPU, 2GB RAM

| *Working Flowchart*

Start → Display Question → Accept Input → Check Answer → Update Score → Result → End

| ***Program Design***

The application uses a Quiz class with methods `__init__`, `start_quiz`, and `show_quiz` for modularity.

| *Code: Init & Questions*

```
self.question=[
    {"Question":"Which is immutable?", "options":["A.Tuple","B.Set","C.Dictionary","D.List"], "answer":"A.
    Tuple"},
    {"Question":"Which data type is used to store text in Python?", "options":["A. int","B. float","C.
    str","D. bool"], "answer":"C. str"},
    {"Question":"Which function is used to take input from the user?", "options":["A. print()","B.
    input()","C. scan()","D. get()"], "answer":"B. input()"}]
```

| *Code: Quiz Logic*

```
def start_quiz(self):  
    for i in self.question:  
        answer=input("Enter  
Answer(A/B/C/D):").upper()  
        if answer in ("A","B","C","D"): break  
        if answer == i["answer"][0]: self.score += 1
```

| *Sample Execution*

System: "Which function is used to take input?"

User: "B"

Output: "Correct Answer!"

| ***Advantages***

- *High Performance & Speed.*
- *High Portability.*
- *Easy Maintainability.*

| *Future Scope*

- *GUI Integration (Tkinter).*
- *Database persistence (MySQL).*
- *User login and timer-based quizzes.*

Conclusion & Thank You

This project demonstrates Python core logic and OOP concepts effectively.

Any Questions?